

IX Open Autumn University Students' Mathematical Competition
OSAM Comp 2026 by FCS HSE

September 13th 2026, 11:00 am – 2:00 pm

Category I (1st year bachelors)

1. A function f has the property that for every real number x in its domain, the number $\frac{1}{x}$ is also in its domain, and

$$f(x) + f\left(\frac{1}{x}\right) = x.$$

What is the largest set of real numbers that can be the domain of f ?

2. Let n and k be positive integers such that $k \leq n$. In how many ways can one choose k integer intervals in the set $\{1, 2, \dots, n\}$ so that the intersection of any two of them is empty?

Note: An *integer interval* is a set consisting of one or several consecutive integers.

3. Let $a, b > 0$. Define the ellipses

$$\mathcal{E}_1 = \left\{ (x, y) \in \mathbb{R}^2 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \right\}, \quad \mathcal{E}_2 = \left\{ (x, y) \in \mathbb{R}^2 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 4 \right\}.$$

Let A be a point on $\mathcal{E}_2$. Let B and C be distinct points on $\mathcal{E}_2$ such that the lines AB and AC are tangent to $\mathcal{E}_1$. Prove that BC is also tangent to $\mathcal{E}_1$ and that the area of triangle ABC does not depend on the choice of the point A .

4. Let p be a prime number. Define

$$A = \{a^2 \bmod p : a \in \mathbb{Z}\}, \quad B = \{b^4 \bmod p : b \in \mathbb{Z}\}.$$

For which p does $A + B = \mathbb{Z}/p\mathbb{Z}$ hold, that is, every residue modulo p is representable as a sum of a square and a fourth power?

5. Let $p(x) \in \mathbb{R}[x]$ be a polynomial such that

$$\deg p \leq 2018 \quad \text{and} \quad |p(x)| \leq \frac{1}{|x - \sqrt{3}|} \quad \text{for all } x \in [-2, 2] \setminus \{\sqrt{3}\}.$$

Prove that

$$|p(\sqrt{3})| \leq 2019.$$